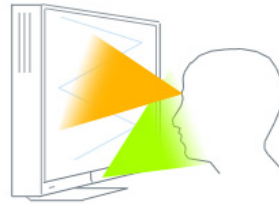
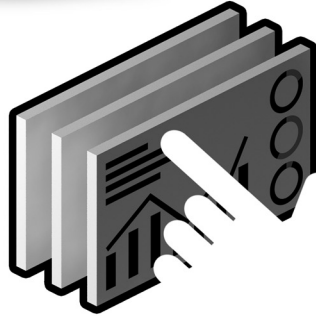
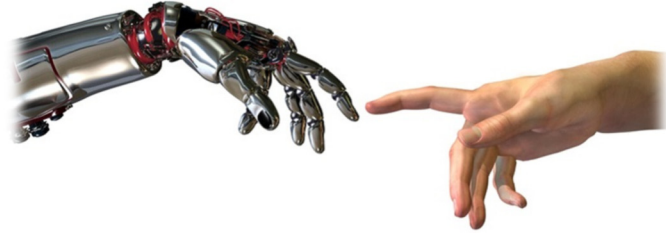


Multimedia, Interaction, Interfaces



What is *multimedia*?

Possible definition (not the only one):

Multimedia is a “presentation” or “interface”, based on the use of the computer, which includes at least two of the following elements:

- Text
- Sound
- Images
- Animations
- Video

Alternative terms:

Hypermedia, mixed media, interactive multimedia, multiple media

Offline multimedia

Conceptually, there are very few differences between online and offline content (in the past, CD/DVD/Blu-ray Disc)

Multimedia resources are still surfed and accessed through “links” (or buttons, or other graphical elements)

However, offline multimedia is less “constrained”

- Inclusion of richer content

Images, animations, videos, and audio with smaller size limits

- More interaction possibilities

Local memory read/write, execution of external applications, use of input devices different from keyboard/mouse/touch, etc.

Interactivity

Possible definition

The communication process between two parts, where each part causes changes in the other, and/or reactions. In Human-Computer communication, interactivity is achieved through the *interface*

Interactivity is essential for multimedia

An alternative definition of *multimedia* considers interactivity. The user provides some kind of input which determines the behavior of the multimedia program/system: according to this definition, a simple presentation containing text and images cannot be considered a multimedia system

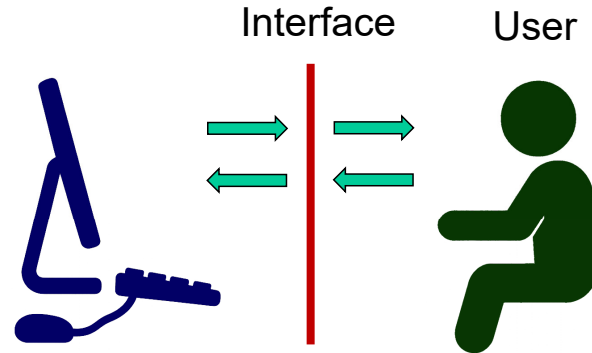
HCI = Human-Computer Interaction

Interface

Possible definition

The “point of interaction” between the human and the computer.

It regards any aspect of the interaction (physical, psychological, temporal, technological, etc.)



Main HCI paradigms

50s - 60s

No real interaction paradigm – Interaction through switches, punched cards...

70s

“Typewriter” paradigm – Interaction through textual commands

80s – 90s

“Desktop” paradigm – Interaction through GUIs (Graphical User Interfaces) based on WIMP (Windows, Icons, Menus, and Pointing devices)

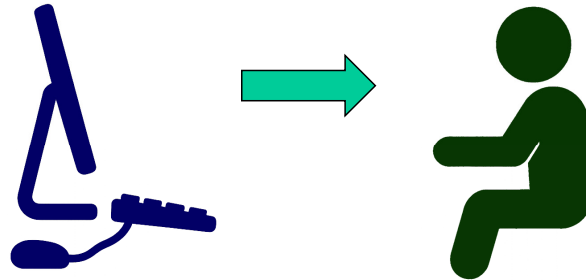
> 2000

“Naturalness” paradigm – Interaction through PUIs (Perceptual User Interfaces)

Multimedia interfaces

Provide different kinds of output (to the user)

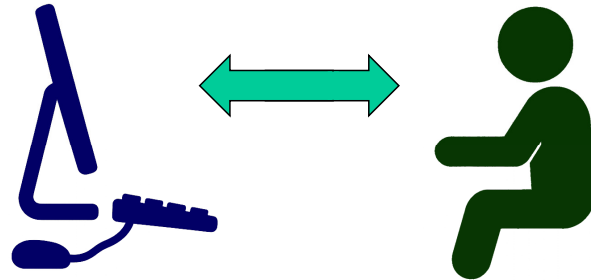
They are focused on the media: text, graphics, audio, video, ...



Multimodal interfaces

Exploit different forms of input (from the user)

- Besides the “ordinary” keyboard and mouse (e.g., voice, gestures, ...)
- Almost always, multimodal interfaces are also multimedia interfaces

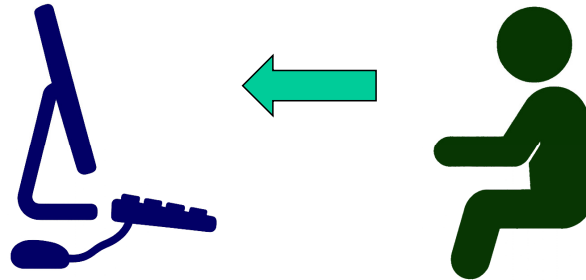


... therefore,
multimedia can
be seen as a
subset of
multimodality

Perceptive interfaces

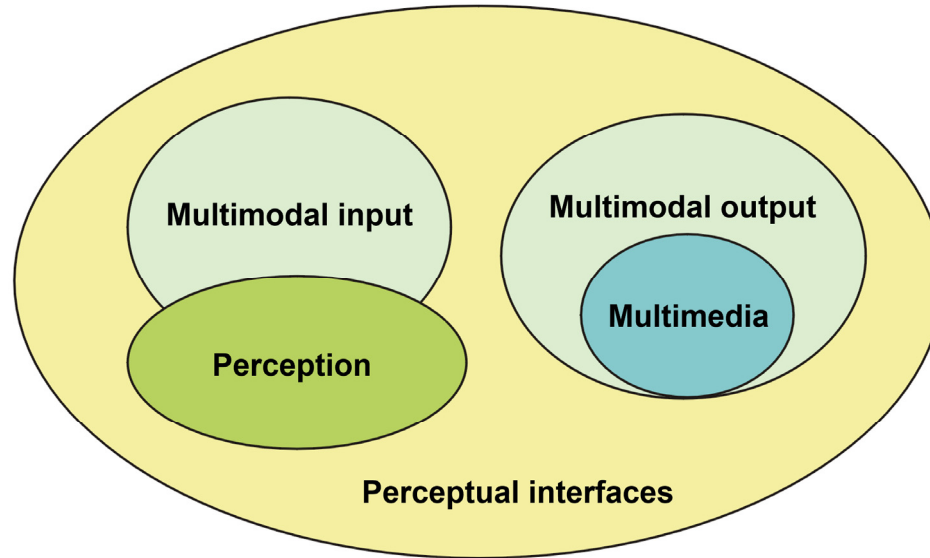
Give the computer “sensory faculties” similar to those of humans

Sight, hearing, ...



Perceptual interfaces

Integrate multimedia, multimodal, and perceptive interfaces



Multimedia evolution

Past

Pictures, cartoons, movies



Present

Multimedia is everywhere (sight and hearing are mainly involved)



Future

Higher interactivity and user control; all five senses are involved;
better sharing through the Internet



Virtual reality...

Synthesis of an artificial environment

Simulated through 3D rendering of virtual objects

Types of virtual reality

- *Non-immersive*

The virtual environment is well separated from the real environment (e.g., video games)

- *Immersive*

The user is fully "immersed" in the virtual environment – e.g., through HMDs (Head Mounted Displays), glasses, gloves, etc.



Beyond reality

... virtual reality...

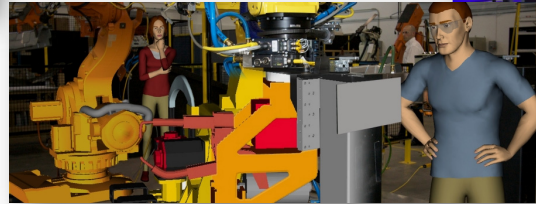
Examples



Virtual aircraft piloting
and car driving



Games



Virtual environments
(with *avatars*)

Beyond reality

... virtual reality

Example: *Meta* devices



Real + Virtual

Augmented Reality...

Virtual elements are added to the real world

Artificial images overlapped with "real images"...

Problems

Correct matching between real and virtual objects

Main difference compared to virtual reality

The user maintains a connection with the real world



Real + Virtual

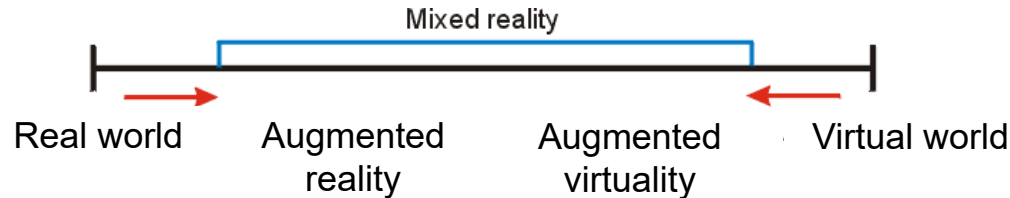
... augmented reality...

Tools for augmented reality

- To acquire information about the real world → cameras, microphones, ...
- For the augmented perception of reality → monitors, HMDs, “glasses”

Mixed reality

Broader concept that takes into account different levels of “reality” or “virtuality”



... augmented reality...

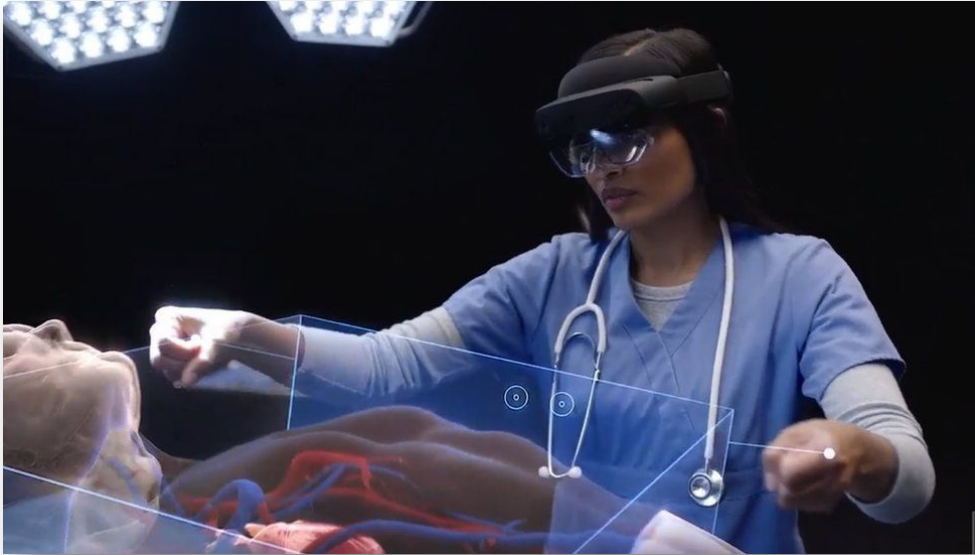
Application areas

- Medical
E.g., surgery
- Military
E.g., additional information for aircraft pilots
- Production, repair, and maintenance of industrial plants
E.g., virtual view of internal components
- Television/advertising
E.g., chroma key
- Furnishings, hairdressing, plastic surgery
E.g., a room with virtual furniture
- Mobile applications for smartphones
Many examples...

Real + Virtual

... augmented reality

Example: *Microsoft HoloLens*



Being “elsewhere”

Telepresence

“Feeling” of being in another place

In a sense, virtual reality and augmented reality are forms of telepresence

Typical use

Exploration of inaccessible places by means of remote-controlled mobile robots endowed with cameras and possibly other sensors



More “natural” HCI?

Gesture interaction...

Several applications

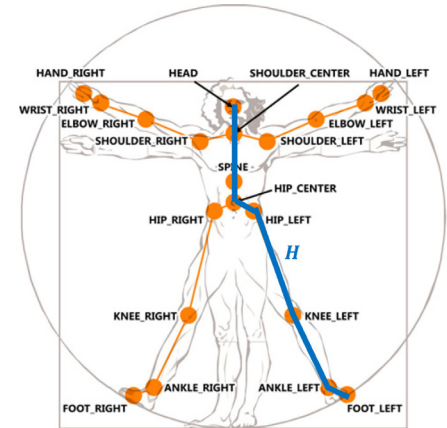
- Videogames
- Mouse replacement
- Interaction at a distance with home appliances (e.g., TVs)
- 3D manipulations
- Biometrics
- ...



More “natural” HCI?

... gesture interaction...

Example: Microsoft *Kinect* (for Windows) - discontinued



More “natural” HCI?

... gesture interaction

Example: *Leap Motion*



Leap Motion, HP Team Up for 3D Motion Control



By Stephanie Miot

April 16, 2013 01:08pm EST

9 Comments

Leap Motion is teaming up with Hewlett-Packard to bring its [3D](#) motion control technology to select HP devices, though no launch timeline or product list were released.

The partnership will begin by bundling the Leap Motion Controller with certain HP products, but that will evolve into the sale of HP devices embedded with Leap Motion's technology.

"Our focus at Leap Motion is to fundamentally improve how people interact with their devices, and offer as many ways as possible to achieve that vision," said Leap CEO Michael Buckwald.



More “natural” HCI?

Eye Tracking...



Kinds of eye trackers

- Remote
- Wearable



More “natural” HCI?

... Eye Tracking

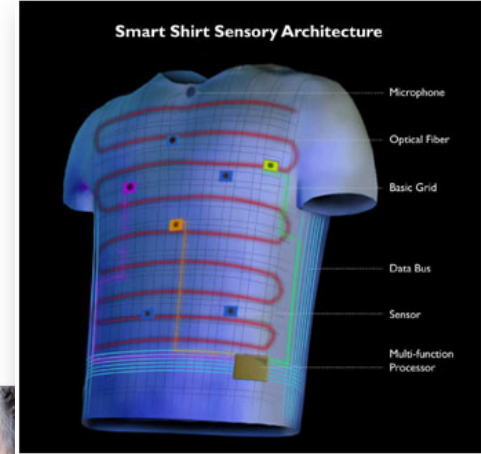


gazepoint



More “natural” HCI?

Wearable computers



More “natural” HCI?

Brain-Computer Interfaces (BCI)

